Exam Date & Time: 25-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

SEVENTH SEMESTER B.TECH END SEMESTER EXAMINATIONS, NOV 2024

Data Structures and Algorithms [ECE 4070]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Instructions to Candidates:
Missing data may be suitably assumed.
Do not use libraries like vector/stack etc.

1) Write an interactive C++ program that performs the following tasks:

A) i) Insert elements into a sorted array of size 10 while maintaining the sorted order.
ii) Merge two sorted arrays into a single sorted array (avoid duplicate values). (4)
Assume data are sorted in ascending order.

B) What do you mean by efficiency of an algorithm? Explain the concept of best case, average case and worst-case time complexity with example. (3)

C) Create a class called COMPLEX which has data members to store real and imaginary part of a complex number. The class also has the functions read_info() to read, dis_info() to display and mul_complex() to multiply two complex numbers. Write an interactive C++ program to add and multiply two complex numbers. (3)

2) Write an interactive C++ program that implements a stack using an array to perform the functions push(), pop(), and peek(), as well as the main function with all necessary error handling. (4)

A)

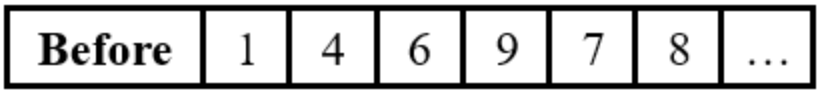
B) Convert the following Infix expression into Prefix form. Show all the steps clearly and also trace the stack content after every step.
(3)
W - (X / Y + (Z / S ^ T) * R) / V.

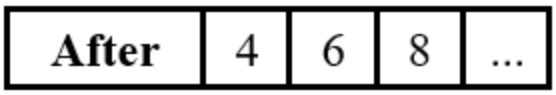
C) What are the limitations of linear queue over circular queue? Perform the following Queue operation using pointers on linear queue and circular queue and draw the queue diagram after each operation: enqueue(11), enqueue(20), enqueue(24), dequeue(), enqueue(60). Assume Queue size as 3 and initially empty. (3)

3) Write an algorithm and interactive program to perform addition of two polynomials and display the result using doubly linked list. (4)

A)

B) Define a class called SLIST which represent singly linked list and contains nodes as shown in Figure (Before). Write an interactive C++ program with member functions to perform the following:





(3)

i) Create a list with 10 nodes

ii) Delete the nodes in the list containing odd data value (refer Figure (After))

iii) Display the list

C) Define a class for a double-linked list in which data to be stored in each node are (a) student name (char array), (b) student marks (int), (c) Subject code(char array). Write an interactive C++program to perform the following functions by handling all error conditions

i) Append N nodes, where N is provided by the user (3)

ii) Update the contents of the Yth node if exist, otherwise display error message

4) What is a height balanced tree? Insert the following nodes one-by-one in the height balanced and balance the tree at every step: 21, 26, 30, 9, 4, 14, 28, 18, 15 (4)

A)

B) A file contains the following symbols with the frequencies as shown.

Symbol	a	e	i	o	u	s	t
Frequency	10	15	12	3	4	13	1

Construct Huffman tree and determine: (3)

i) Huffman Code for each character

ii)Average code length

iii)Length of Huffman encoded message (in bits)

C) Consider a tree having pre-order and in-order traversal as follows. Draw the tree and write down its post-order traversal. (3)

In-order: Z, A, Q, P, Y, X, C, B

Pre-order: Q, A, Z, Y, P, C, X, B

5) Construct a directed graph for the given adjacency matrix.

A)

	1	2	3	4	5	6
1	0	1	0	1	0	0
2	0	0	1	0	0	0
3	0	0	0	1	1	0
4	0	0	0	0	0	0
5	0	1	0	1	0	0
6	1	0	1	1	0	0

(4)

Determine Depth First Search traversal output of the graph with node 1 as source. Clearly show the status after each iteration.

B) Write a C++ program to perform binary search on ***n*** data elements for a given key ***k***. What are the best-case and worst-case time complexities. (3)

C) Discuss the heap sort method of sorting a list. What is its time complexity? Show its operation on the list containing elements: 42, 32, 48, 20, 58, 53, 75, 53, 97 (3)

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